in DSP. If you can't grasp one, maybe the other will help. In BASIC, the % character at the end of a variable name indicates it is an integer. All other variables are floating point. Chapter 4 discusses these variable types in detail.

This method of calculating the mean and standard deviation is adequate for many applications; however, it has two limitations. First, if the mean is much larger than the standard deviation, Eq. 2-2 involves subtracting two numbers that are very close in value. This can result in excessive round-off error in the calculations, a topic discussed in more detail in Chapter 4. Second, it is often desirable to recalculate the mean and standard deviation as new samples are acquired and added to the signal. We will call this type of calculation: **running statistics.** While the method of Eqs. 2-1 and 2-2 can be used for running statistics, it requires that *all* of the samples be involved in each new calculation. This is a very inefficient use of computational power and memory.

A solution to these problems can be found by manipulating Eqs. 2-1 and 2-2 to provide another equation for calculating the standard deviation:

EQUATION 2-3
Calculation of the standard deviation using running statistics. This equation provides the same result as Eq. 2-2, but with less round-off noise and greater computational efficiency. The signal is expressed in terms of three accumulated parameters: *N*, the total number of samples; *sum*, the sum of these samples; and *sum of squares*, the sum of the squares of the samples. The mean and standard deviation are then calculated from these three accumulated parameters.

$$\sigma^2 = \frac{1}{N-1}\left[\sum_{i=0}^{N-1} x_i^2 - \frac{1}{N}\left(\sum_{i=0}^{N-1} x_i\right)^2\right]$$

or using a simpler notation,

$$\sigma^2 = \frac{1}{N-1}\left[sum\ of\ squares - \frac{sum^2}{N}\right]$$

While moving through the signal, a running tally is kept of three parameters: (1) the number of samples already processed, (2) the sum of these samples, and (3) the sum of the squares of the samples (that is, square the value of each sample and add the result to the accumulated value). After any number of samples have been processed, the mean and standard deviation can be efficiently calculated using only the current value of the three parameters. Table 2-2 shows a program that reports the mean and standard deviation in this manner as each new sample is taken into account. This is the method used in hand calculators to find the statistics of a sequence of numbers. Every time you enter a number and press the Σ (summation) key, the three parameters are updated. The mean and standard deviation can then be found whenever desired, without having to recalculate the entire sequence.